Question 4

(a)

$$\begin{tikzcd} T \arrow[r] \arrow[rr, bend right] & M \arrow[r] & Y \end{tikzcd}$$

$$Y = 2(\alpha T + \varepsilon_M) + (3-\alpha^2)T + \varepsilon_Y$$
$$= (2\alpha + 3 - \alpha^2)T + 2\varepsilon_M + \varepsilon_Y$$

(b) $\because E[\varepsilon_M] = 0$

$E[\varepsilon_Y] = 0$

$\therefore E[Y \mid do(T=1)] = 3 + 2\alpha - \alpha^2$

$E[Y \mid do(T=0)] = 0$

$\tau(\alpha) = E[Y \mid do(T=1)] - E[Y \mid do(T=0)]$
$= 3 + 2\alpha - \alpha^2$

(c) $\tau(\alpha) = 3 + 2\alpha - \alpha^2$

$\tau'(\alpha) = 2 - 2\alpha$

$\tau''(\alpha) = -2 < 0$

$\therefore$ when $\tau'(\alpha) = 0$, $\alpha = 1$

$\tau(\alpha)$ reaches the maximum.

$\max_{\alpha} \tau(\alpha) = \tau(1) = 4$

(d) Increasing α strengthens the positive mediated effect through job-readiness M, but it also makes the direct effect of the treatment on Y similar due to the term $(3-a^2)T$. For large α, the negative quadratic term dominates.

(e) The optimal value of a is 1.5 under the constraint of $[1.5, 2]$. ~~since~~ ~~$\tau(a)$~~ $\tau(a)$ is monotonically decreasing on $a \in [1.5, 2]$, since $\tau'(\alpha) < 0$ for all ~~$\alpha > 0$~~ $\alpha > 1$

(f) when $\alpha = 1$,

$M = T + \varepsilon_M$

$Y = 2M + 2T + \varepsilon_Y$

$\because T = 1, M = 1.5, Y = 4$

$\therefore M = T + \varepsilon_M \quad Y = 2M + 2T + \varepsilon_Y$

$\varepsilon_M = 0.5 \quad \varepsilon_Y = -1$

$\therefore$ when $T = 0$,

$M(0) = 0 + \varepsilon_M = 0.5$

$Y(0) = 2M(0) + 2(0) + \varepsilon_Y = 0$

(b) continue

The customer should continue collecting ~~off~~.

$$(R_1+R_2) - cE[T_2] \geq R_1 - cE[T_1]$$

$$\begin{cases} E[T_2] \approx 19.0414 \\ E[T_1] \approx 11.4167 \end{cases}$$

i.e. $R_2 \geq c(E[T_2] - E[T_1])$

$$R_2^{min} \approx 7.6247\, c$$

Question 2.

(a). (i) $A \not\perp C \mid S, Y$

For the path: $A \to D \to Y \leftarrow P \leftarrow C$

Y is the collider but is controlled now.

(ii) ~~$A \not\perp$~~ $Y \perp A \mid D, P$

All paths are blocked

(iii) $D \not\perp S \mid A, P, C$

~~$\not\perp$~~ The path $D \leftarrow S$ is always open.

~~(b)~~ (c) Since D is the only parent of Y, the effect of ~~intervening~~ intervening on A must operate through the induced distribution of D.

$$\therefore E[Y \mid do(A=a)] = \sum_d E[Y \mid do(A=a), D=d] P(D=d \mid do(A=a))$$

$$\because E[Y \mid do(A=a), D=d] = E[Y \mid do(D=d)]$$

$$\therefore E[Y \mid do(A=a)] = \sum_d E[Y \mid do(D=d)] P(D=d \mid do(A=a))$$

$\therefore$ ~~$\exists$~~ There exists at least one d^* such that

$$E[Y \mid do(D=d^*)] \geq \sum_d E[Y \mid do(D=d)] P(D=d \mid do(A=a))$$

$$\therefore E[Y \mid do(D=d^*)] \geq E[Y \mid do(A=a)]$$

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